

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
= 1	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — S(R,θ)

Far-field E,H phasors small l ≪ λ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$	
Avg power density l/λ < 1/50; far field $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$ (W/m ²)	
<i>Equivalent:</i> $S = S_{\text{max}} \sin^2 \theta$ with $S_{\text{max}} = \eta_0 k^2 I_0^2 l^2 / (32\pi^2 R^2)$ at $\theta = \pi/2$.	
l: rod length (m)	I ₀ : peak current (A)
R: obs. distance (m)	θ: angle from dipole axis (rad)
k = 2π/λ (rad/m)	η ₀ = 120π ≈ 377 Ω
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — S(θ)

Avg power density any l; far field $S(R, \theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$ (W/m ²)	
<i>At θ = 0, π: bracket is 0/0. L'Hôpital ⇒ S = 0 (no axial radiation). TI-36X alt: evaluate bracket at θ = 0.001 rad → tiny → 0.</i>	
Normalized pattern dimensionless $F(\theta) = S(\theta) / S_{\text{max}}$	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by π/180 or 180/π $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$						
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ
0	0	0	1	0	1	
30	π/6	1/2	√3/2	1/4	3/4	
45	π/4	√2/2	√2/2	1/2	1/2	
60	π/3	√3/2	1/2	3/4	1/4	
90	π/2	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$	
Normalize $F_a^{\text{norm}} = F_a / N^2$	
HPBW (broadside, uniform) use Nd, not (N−1)d $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$	
Electronic steering (scan to θ ₀) replace γ with γ' = kd(cos θ − cos θ₀) $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elr}}$); $\theta_0 = \frac{\pi}{2}$: broadside, 0: end-fire	
Frequency scan feed-line incr. Δℓ = n₀ λ₀ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$	
<i>Grating lobes appear if d > λ. Avoid.</i>	

BEAM PARAMETERS — β, Ω_p, D

Normalize always start here $F(\theta) = S(\theta) / S_{\text{max}}$ (unitless) Beamwidth β at threshold X Set $F(\theta_X) = X$, solve for θ _X ; β = 2θ _X (symmetric)		
Threshold	X = 10 dB/10	θ _X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any F(θ _X)	any X	$F(\theta_X) = X$
Pattern solid angle Ω _p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)		
<i>No φ dep.: Ω_p = 2π ∫₀^π F(θ) sin θ dθ.</i> <i>θ is the integration variable — DON'T plug in a number for θ (e.g. θ_{1/2}). Integrate over θ ∈ [0, π].</i>		
Pattern F(θ)	Ω _p (sr)	
Isotropic (F = 1)	4π ≈ 12.57	
Hertzian sin ² θ	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss e ^{−aθ²}	π/a = πθ ₂ ² /2 / ln 2	
2-axis HPBW (aperture)	β _{xz} · β _{yz}	
Other (use integral)	numerical	

TI-36X: [2nd] [d/dx] for $\int \square$, RAD mode, multiply by 2π.

Directivity D always D = 4π/Ω_p; common shortcuts below $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$		
WORD TRAP: “direction” = θ _{max} (rad); “directivity” = D (unitless).		
Gain (if asked) ξ = radiation efficiency (unitless), 0 < ξ ≤ 1 $G = \xi D$ (unitless) $G_{\text{dB}} = 10 \log_{10} G$ (dB)		
Lossless / ideal antenna ⇒ G = D.		
$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$; $R_{\text{rad}} = 2P_{\text{rad}} / I_0^2$ (Ω); $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$.		

COMMON DIPOLE PARAMETERS

Lossless (ξ = 1): G = D.				
Type	l	D	D _{dB}	R _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< λ/50	1.5	1.76	80π ² (l/λ) ²
Half-wave	λ/2	1.64	2.15	73.2
Monopole	λ/4 (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole (λ/4 above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.				

ANTENNA AREAS (A_e, A_p)

Effective area antenna's RX cross section; D from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter d; l see table above $A_p = l d$ (m ² , any dipole)	
Inverse: D from A _e $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire: A_e ≫ A_p (e.g., Hertzian A_e = 3λ²/8π ≈ 0.119λ²).</i>	

FRISIS TRANSMISSION FORMULA

For each antenna's G: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (G = D, lossless). Else (dB/linear value given) → use as given (convert dB → linear).
Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$
 $S_r = \frac{G_t P_t}{4\pi R^2}$ (W/m²); $P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$ (W)

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.
Recipe 1. λ = c/f. 2. Convert any dB gain → linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get G_t, G_r by antenna type: dipole → COMMON DIPOLE PARAMS; aperture → APERTURE box; arbitrary F → BEAM PARAMS. 4. Plug into Friis.
Use linear G, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity (G, D, P_r/P_t, SNR, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power ∝ E²).

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23}$ J/K; T _{sys} system noise temp (K); B receiver bandwidth (Hz).	
Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$	

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); A_e effective area (m², = λ²D/4π); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ, d aperture side / diameter (m).
Far-field req.: R ≥ 2d²/λ (d = largest aperture dim.).

Pattern (rect., uniform illum.) peak at θ = 0 (broadside)
 $F(\theta) = \text{sinc}^2 \left(\frac{\pi \ell_x \sin \theta}{\lambda} \right)$; $S_0 = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}$

Directivity (any shape, uniform illum.)
 $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ $A_e \approx A_p$
Directivity from beamwidths any aperture
 $D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$

HPBW and D by aperture shape uniform illum.; β in rad			
Shape	A _p	HPBW (rad)	D (unitless)
Rect. ℓ _x × ℓ _y		$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ ²	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	πd ² /4	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$

Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$
 $D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$
 $\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$

If a parameter is unchanged, its ratio = 1. Examples: double area → D ×2, β ×1/√2; double freq → D ×4, β ×1/2.

DIPOLE

l — antenna length (m)
λ = c/f (m); c = 3 × 10 ⁸ m/s
I ₀ — peak current (A)
k = 2π/λ (rad/m)
R — obs. dist. (m); θ from axis
η ₀ = 120π ≈ 377 Ω
S(R, θ) — pwr density (W/m ²)

BEAM

F(θ) = S/S _{max} (unitless)
a — coef. on θ ² in Gaussian
β — HPBW (rad or °)
Ω _p — pattern solid angle (sr)
D = 4π/Ω _p — directivity
ξ radiation eff. (unitless); G = ξD
D _{dB} = 10 log ₁₀ D; D = 10 ^{D_{dB}/10}

FRISIS / dB

P _t — power transmitted (W)
P _r — power received (W)
G _t — transmit gain (linear)
G _r — receive gain (linear)
S _r = G _t P _t / (4πR ²) (W/m ²)
P _r = G _t G _r (λ/4πR) ² P _t
G = 10 ^{G_{dB}/10}
3 dB = ×2, 10 dB = ×10
P _N = k _B T _{sys} B (W)

APERTURE

A _p — physical area (m ²)
A _e = λ ² D/(4π) (m ²)
A _e ≈ A _p for aperture
D ≈ 4πA _p /λ ² (unitless)
β ≈ 0.88λ/ℓ (rect/sq.)
β ≈ 1.02λ/d (circular)
2A _p ⇒ 2D, β/√2

ARRAY

N — number of elements
d — spacing (m)
γ = kd cos θ
F _a = sin ² (Nγ/2) / sin ² (γ/2)
F _a ^{max} = N ² at θ = π/2
F _a ^{norm} = F _a /N ²
β ≈ 0.88λ/(Nd) (broadside)